

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Prompt Engineering Challenge

Course Code:	CSA6502	Course Title:	Generative AI and Large Language Models
CO Assessed:	CO2 – Precision (BL1, BL2)	Assessment:	Assessment Tool 3 – Prompt Engineering Challenge
Weightage:	40% of CIA	Total Marks:	100 (2 Question)
CO1 Statement:	<i>Apply prompt engineering techniques and utilize pre-trained Large Language Models through modern AI frameworks and APIs to solve real-world problems.(BL1, BL2)</i>		
Student Name:	S. Muskan	Reg. No.:	192472182
Section / Batch:	BE-CSE-AI	Date:	29-07-2026

Instructions to Students

- Answer 2 questions. Each question carries 50 Marks. Total: 100 Marks.
- Write clear, well-structured prompts and explanations where asked.
- Use your own original examples — do not copy from classmates or the internet verbatim.
- Where a diagram or worked example is requested, label it clearly.
- Illustrate prompting techniques (zero-shot, few-shot, chain-of-thought, role/system prompting, etc.) with concrete examples, not just definitions. Time allotted: 30 mins

Code of Conduct

I S.Muskan(192472182) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: S. Muskan

SECTION A – PROMPT ENGINEERING CHALLENGE

Q8. Self-Consistency / Verification Prompt

Write a prompt that asks the model to solve a problem, then re-check its own answer in a second pass and flag if it finds an error.

Q11. ReAct Prompting (Reason + Act)

Write a ReAct-style prompt that makes the model alternate between Thought:, Action:, and Observation: steps to answer a question that requires looking something up (e.g., "What's the population of the capital of the country that won the 2022 World Cup?"), simulating a tool-use loop.

Q8. Self-Consistency / Verification Prompt

Aim:

To create a prompt that makes an AI solve a problem and then verify its own answer in a second pass.

Algorithm:

1. Accept the problem statement.
2. Generate the initial solution.
3. Perform a second verification pass.
4. Compare both results.
5. Report corrections if any, otherwise confirm verification.

Python Code:

```
Googleapi.py - C:/Users/vishn/AppData/Local/Programs/Python/Googleapi.py (3.10.11)
File Edit Format Run Options Window Help
from google import genai

# Replace with your Gemini API Key
client = genai.Client(api_key="AQ.Ab8RN6IDhZF_HxHVL29ChuD3E_zxZz7czS6oXdjlwSahC5z_QQ")

prompt = """
You are a careful problem-solving assistant.

Task:
Solve the following problem step by step.
After obtaining your final answer, perform a second verification pass.

Rules:
1. Solve the problem step by step.
2. Give the final answer.
3. Re-check every calculation and logical step.
4. If you find any mistake, explain it and correct it.
5. If no errors are found, print:
    "Verification successful. No errors detected."

Problem:
If a train travels 180 km in 3 hours, what is its average speed?
"""

response = client.models.generate_content(
    model="gemini-2.5-flash",
    contents=prompt
)

print(response.text)
```

Output:

```
IDLE Shell 3.10.11
File Edit Shell Debug Options Window Help
===== RESTART: C:/Users/vishn/AppData/Local/Programs/Python/Googleapi.py =====
To solve this problem, we need to find the average speed of the train.

**Step-by-step derivation:**

1.  **Identify the given information:**
    *   Distance traveled = 180 km
    *   Time taken = 3 hours

2.  **Recall the formula for average speed:**
    The average speed is calculated by dividing the total distance traveled by the total time taken.
    Average Speed = Total Distance / Total Time

3.  **Substitute the given values into the formula:**
    Average Speed = 180 km / 3 hours

4.  **Perform the calculation:**
    Average Speed = 60 km/h

**Final Answer:**
The average speed of the train is 60 km/h.

**Second Verification Pass:**

*   **Problem understanding:** The question asks for average speed given distance and time. This is correctly understood.
*   **Formula used:** The formula for average speed (Distance / Time) is correct.
*   **Values substituted:** Distance = 180 km and Time = 3 hours are correctly substituted.
*   **Calculation:** 180 divided by 3 is indeed 60. The arithmetic is correct.
*   **Units:** The units are kilometers (km) for distance and hours (h) for time, so the speed is correctly expressed in kilometers per hour (km/h).
*   **Logical flow:** The steps follow a logical progression from identifying inputs to applying the formula and performing the calculation.

No errors were found during the verification pass.

Verification successful. No errors detected.
>>>
```

LLM Response:

Prompt:

You are a careful problem-solving assistant. Task: Solve the following problem step by step. After obtaining your final answer, perform a second verification pass. Rules:

1. Solve the problem step by step.
2. Give the final answer.
3. Re-check every calculation and logical step.
4. If you find any mistake, explain it and correct it.
5. If no errors are found, print: "Verification successful. No errors detected."

Problem: If a train travels 180 km in 3 hours, what is its average speed?

Response:

To solve this problem, we need to find the average speed of the train.

Step-by-step derivation:

1. Identify the given information:
Distance traveled = 180 km
Time taken = 3 hours

2. Recall the formula for average speed:

The average speed is calculated by dividing the total distance traveled by the total time taken.

Average Speed = Total Distance / Total Time

3. Substitute the given values into the formula:

Average Speed = 180 km / 3 hours

4. Perform the calculation:

Average Speed = 60 km/h

Final Answer:

The average speed of the train is 60 km/h.

Second Verification Pass:

Problem understanding: The question asks for average speed given distance and time. This is correctly understood.

Formula used: The formula for average speed (Distance / Time) is correct.

Values substituted: Distance = 180 km and Time = 3 hours are correctly substituted.

Calculation: 180 divided by 3 is indeed 60. The arithmetic is correct.

Units: The units are kilometers (km) for distance and hours (h) for time, so the speed is correctly expressed in kilometers per hour (km/h).

Logical flow: The steps follow a logical progression from identifying inputs to applying the formula and performing the calculation.

No errors were found during the verification pass.

Verification successful. No errors detected.

Result:

The model first solves the problem and then re-checks its answer, correcting any detected errors or confirming that the answer is verified.

Q11. ReAct Prompting (Reason + Act)

Aim:

To create a ReAct-style prompt that alternates between reasoning and actions for tool-assisted problem solving.

Algorithm:

1. Receive the user question.
2. Generate a Thought step.
3. Perform an Action (lookup/search).
4. Record the Observation.
5. Repeat until enough information is collected.
6. Produce the Final Answer.

Python Code:

```
Googleapi.py - C:/Users/vishn/AppData/Local/Programs/Python/Googleapi.py (3.10.11)
File Edit Format Run Options Window Help
from google import genai

# Replace with your Gemini API Key
client = genai.Client(api_key="AQ.Ab8RN6IDhZF_HxHVL29ChuD3E_zxZz7czS6oXdjlwSahC5z_QQ")

prompt = """
You are an AI assistant that follows the ReAct (Reason + Act) prompting framework.

Instructions:
1. Alternate between the following sections:
    Thought:
    Action:
    Observation:
2. Continue until you have enough information.
3. End with:
    Final Answer:
4. If a real search tool is unavailable, simulate the lookup using your knowledge.

Question:
What is the population of the capital of the country that won the 2022 FIFA World Cup?
"""

response = client.models.generate_content(
    model="gemini-2.5-flash",
    contents=prompt
)

print(response.text)
```

Output:

```
Python 3.10.11 (tags/v3.10.11:7d4cc5a, Apr 5 2023, 00:38:17) [MSC v.1929 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: C:/Users/vishn/AppData/Local/Programs/Python/Googleapi.py =====
Action: Search
Query: "who won 2022 FIFA World Cup"Observation: Argentina won the 2022 FIFA World Cup.Thought: I have identified the country. Now I need to find its capital city.Action: Search
Query: "capital of Argentina"Observation: The capital of Argentina is Buenos Aires.Thought: I have identified the capital city. Now I need to find its population.Action: Search
Query: "population of Buenos Aires"Observation: The population of Buenos Aires (City of Buenos Aires) is approximately 3 million (2.89 million as of the 2010 census, and estimated around 3.1 million in 2022). The larger metropolitan area (Greater Buenos Aires) has a population of around 15 to 16 million.

The question asks for the "population of the capital", which generally refers to the city proper, not the entire metropolitan area unless specified. I will use the figure for the City of Buenos Aires.Final Answer: The population of Buenos Aires, the capital of Argentina, is approximately 3.1 million.
>>>
```

LLM Response:

Prompt:

You are an AI assistant that follows the ReAct (Reason + Act) prompting framework.

Instructions:

1. Alternate between the following sections:

Thought:

Action:

Observation:

2. Continue until you have enough information.

3. End with: Final Answer:

4. If a real search tool is unavailable, simulate the lookup using your knowledge.

Question: What is the population of the capital of the country that won the 2022 FIFA World Cup?

Response:

Action: Search

Query: "who won 2022 FIFA World Cup"Observation: Argentina won the 2022 FIFA World Cup.

Thought: I have identified the country. Now I need to find its capital city.

Action: Search

Query: "capital of Argentina"Observation: The capital of Argentina is Buenos Aires.

Thought: I have identified the capital city. Now I need to find its population.

Action: Search

Query: "population of Buenos Aires"Observation: The population of Buenos Aires (City of Buenos Aires) is approximately 3 million (2.89 million as of the 2010 census, and estimated around 3.1 million in 2022). The larger metropolitan area (Greater Buenos Aires) has a population of around 15 to 16 million.

The question asks for the "population of the capital", which generally refers to the city proper, not the entire metropolitan area unless specified. I will use the figure for the City of Buenos Aires. Final Answer: The population of Buenos Aires, the capital of Argentina, is approximately 3.1 million.

Result:

The model follows the Thought → Action → Observation cycle before producing the final answer, improving reasoning transparency and simulated tool use.

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technique Selection	30%	Correctly applies zero-shot/one-shot/few-shot & CoT technique appropriate to each task	Applies techniques with minor mismatch to task	Limited understanding of which technique to use	Techniques applied incorrectly or not evident
Output Effectiveness	25%	Prompts produce accurate, task-appropriate outputs across all assigned tasks	Prompts work well for most tasks	Prompts work for only a few tasks	Prompts are largely ineffective
Prompt Craftsmanship	20%	Prompts are clear, specific, and reusable	Prompts are mostly clear	Prompts are vague or inconsistent	Prompts are poorly constructed
Prompt-Output Documentation	15%	Well-documented prompt-output pairs with clear rationale	Documented with minor gaps	Minimal documentation of prompts/outputs	No rationale provided for prompt choices
Design Rationale	10%	Clearly explains design choices and iterations made	Explains most design choices	Limited explanation of process	No explanation of process provided

Marks Evaluation:

Question No.	Technique Selection (30%)	Output Effectiveness(25%)	Prompt Craftsmanship (20%)	Prompt-Output Documentation (15%)	Design Rationale(10%)	100
Q1						
Q2						
Overall Total (100)						

Faculty In-charge

Course Coordinator

Head of Department

Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____